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Title: Windows Server Backup and Recovery

1. Open Windows Server Backup on your training VM and perform a one-time full backup of the C: drive. Take a screenshot of the confirmation screen after completion.

⇒ Steps

Step 1

Open Server Manager.

Step 2

Click:

Tools → Windows Server Backup

Note: If Windows Server Backup is not installed:

Go to Manage → Add Roles and Features

Click Features

Check Windows Server Backup

Click Install

Step 3

In Windows Server Backup, click:

Local Backup

Step 4

Click:

Backup Once...

Step 5

Select:

Different options

Click Next.

Step 6

Choose:

Custom

Click Next.

Step 7

Click:

Add Items

Select:

Local Disk (C:)

Click OK.

Step 8

Click Next.

Step 9

Choose the backup destination:

Local Drive (Example: E:)

External Hard Disk

Another Volume

Click Next.

Step 10

Review the settings.

Click:

Backup

Step 11

Wait until the backup finishes.

You should see:

The backup operation completed successfully.

2. Configure a scheduled incremental backup in Windows Server

Backup to run daily at 8:00 PM for the folder containing your music or video files (e.g., D:\Media). Document the steps you followed and the schedule settings you chose.

⇒ Steps

Step 1

Open:

Server Manager → Tools → Windows Server Backup

Step 2

Click:

Local Backup

Step 3

Click:

Backup Schedule...

Step 4

Click Next.

Step 5

Select:

Custom

Click Next.

Step 6

Click:

Add Items

Select:

D:\Media

Click OK.

Step 7

Click Next.

Step 8

Choose:

Once a day

Set the time to:

8:00 PM

Step 9

Click Next.

Step 10

Select the destination:
Dedicated backup disk
Another volume
Remote shared folder
Choose the option available in your lab.
Step 11
Click:
Finish
Step 12
Click:
Yes
The backup schedule is now configured.

3. Create a comparison table in a Word or Excel file showing the differences between full, incremental, and differential backups, including how much storage and time each method might use over a week if you back up a 10GB folder daily and only 1GB changes per day.
Hint: Use simple numbers and explain your reasoning for each backup type.

⇒ **Explanation**

Full Backup

Copies all files every day.
Uses the most storage.
Takes the longest time.
Easy to restore.

Incremental Backup

Copies only files changed since the last backup.
Uses the least storage.
Fastest backup.
Restore requires the full backup plus all incremental backups.

Differential Backup

Copies all changes since the last full backup.
Uses more storage than incremental but less than full.
Faster restore than incremental.

Reasoning

Since:

Initial data = **10 GB**

Daily changes = **1 GB**

Full Backup

$10 \times 7 = 70 \text{ GB}$

Incremental

$10 + (1 \times 6)$

= 16 GB

Differential

$$10 + 1 + 2 + 3 + 4 + 5 + 6$$

= 31 GB

4. List three backup best practices for Windows Server that would be important for a Flipkart-style e-commerce website, and explain why each is critical for minimizing downtime or data loss.

⇒ 1. Perform Daily Incremental Backups

Reason:

Incremental backups save only changed data, reducing backup time and storage while protecting daily transactions.

⇒ 2. Keep Backups on a Separate Storage Device or Cloud

Reason:

If the main server fails due to hardware failure, ransomware, or accidental deletion, the backup remains safe and can be used to restore services quickly.

⇒ 3. Regularly Test Backup Restoration

Reason:

A backup is only useful if it can be restored successfully. Regular testing ensures data integrity and reduces downtime during disasters.

Benefits for an E-commerce Website

Best Practice	Benefit
Daily Incremental Backup	Protects daily orders and customer data
Separate Backup Storage	Prevents data loss during server failure
Regular Restore Testing	Ensures quick recovery and minimizes downtime

